

Rajesh Phulwaria

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GitHub — LinkedIn — Portfolio

Profile

AI/ML-focused BCA student interested in understanding intelligent systems from first principles. Experienced in mathematical analysis of neural networks, PyTorch/TensorFlow implementations, Transformer architectures, LLM fundamentals, and Generative AI systems. Published research on gradient flow and neural network optimization, with a focus on translating theoretical understanding into practical AI systems.

Technical Skills

Languages: Python, C++, SQL, Java (Beginner), JavaScript (Beginner)

Machine Learning: Regression, Classification, Feature Engineering, Natural Language Processing, EDA, Model Selection, Cross-Validation

Deep Learning: PyTorch, TensorFlow, CNNs, LSTMs, Encoder-Decoder, Backpropagation, Gradient Flow, Optimization, Model Evaluation

LLM / Generative AI: Transformers, Self-Attention, Positional Encoding, BPE Tokenization, Embeddings, LLMs, Prompt Engineering, Context Engineering, Function Calling, Hugging Face, Gemini API

Tools / Research: NumPy, Pandas, Scikit-Learn, Git, FastAPI, Docker, Linux, Experiment Design, Literature Reading, Reproducible Experiments, Scientific Documentation

Research & AI Projects

Niyamit — Legal Metrology Compliance System [GitHub](#) — Team Project, SIH 2026 PS 26034

- Contributed as a team member to the development of **Niyamit**, a software system designed to verify packaged-commodity declarations against the **Legal Metrology (Packaged Commodities) Rules, 2011** and the Legal Metrology Act, 2009.
- Contributed to the **OCR, structured field extraction, and deterministic rule-verification pipeline** for checking declarations such as MRP, net quantity, manufacturer details, country of origin, and consumer-care information.
- Worked on translating statutory requirements into machine-checkable compliance rules, including unit validation, declaration presence, measurement-related checks, and violation identification.
- Collaborated with the team on system architecture, testing, compliance workflows, and integration of the backend processing pipeline with the application interface.

Patient Data Management System [GitHub](#) — FastAPI, Streamlit, Docker, AWS EC2

- Developed a full-stack patient data management system with RESTful FastAPI endpoints, Pydantic-based validation, CRUD operations, patient search, sorting, and automatic BMI calculation.
- Built a Streamlit dashboard integrated with the FastAPI backend for interacting with and managing patient records through HTTP/JSON APIs.
- Containerized the application using Docker, published the image to Docker Hub, and deployed the frontend and backend on an AWS EC2 Linux server.
- Live Preview:** Streamlit Dashboard: 16.178.42.216:8501/ — FastAPI Swagger API: 16.178.42.216:8000/docs

Research Paper Summarizer [GitHub](#) — Python, LangChain, Groq

- Built an interactive AI-powered research paper summarizer using LangChain, Groq, and GPT-OSS-120B, supporting configurable explanation style and response length.
- Used LangChain's **PromptTemplate** for dynamic prompt construction and Groq for fast LLM inference; designed the project for future extension into a RAG-based paper assistant.

GPT-Style Tokenizer from Scratch [GitHub](#) — Python, BPE

- Implemented a byte-level BPE tokenizer with pair-frequency analysis, iterative merging, vocabulary construction, and encode/decode pipelines.
- Trained on a diverse corpus and analyzed how merge count affects vocabulary size, tokenization granularity, and sequence length.

Transformer from Scratch [GitHub](#) — NumPy, PyTorch

- Implemented core Transformer components from first principles, including self-attention, positional encoding, feed-forward networks, masking, and normalization.
- Studied tensor transformations and mathematical operations underlying modern Transformer architectures.

Jarvis – LLM Powered AI Assistant [GitHub](#) — Python, Gemini API

- Built an AI assistant supporting contextual conversations, speech recognition, browser automation, and real-time information retrieval.
- Designed a modular architecture for integrating LLM capabilities and external tools.

Research Publication

A Comparative Analysis of Gradient Flow and Vanishing Gradient Effects in Deep Neural Networks
Zenodo, 2026

- Investigated gradient propagation across deep neural networks and analyzed activation functions affecting optimization stability and learning dynamics.
- DOI: 10.5281/zenodo.19351203

Open Source

fraction-raj (PyPI)

Python

Published an open-source Python package implementing rational arithmetic using Python's data model and operator overloading.

Education

Bachelor of Computer Applications (BCA)

2024–2027

Aishwarya College, Rajasthan

Research Experience

Data Science Intern – Cognifyz Technologies

- Built machine learning pipelines involving preprocessing, feature engineering, experimentation, and model evaluation on real-world datasets.

Python Programming Intern – InternsForge

- Developed Python applications while strengthening software engineering and algorithmic problem-solving skills.